



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Divya Konar
Roll No. :	20
Experiment No.:	3
Title of the Experiment:	Implement Correlation Analysis, Distance Measures, and Data Preprocessing Techniques
Date of Performance:	05/08/26
Date of Submission:	05/08/26

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : **Dr. Tatwadarshi P.N.**

Signature :

Date :



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Aim: To implement correlation analysis, distance measures, and basic data preprocessing techniques on the Pima Indians Diabetes Dataset to identify relationships between variables and prepare data for further analysis.

Objectives

After completing this experiment, the student will be able to:

1. Apply correlation and distance measures to analyze relationships and similarities among data observations.
2. Implement basic data preprocessing techniques to prepare a dataset for statistical analysis and machine learning.

Tools Required

- Python 3.x
- Google Colab / Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains medical diagnostic information for **768 female patients** and is used to study factors associated with diabetes.

The dataset contains numerical attributes and a binary outcome variable indicating whether the patient is diabetic.

Procedure

1. Load the Pima Indians Diabetes Dataset.



2. Identify missing or invalid values, particularly zero values in medical attributes where zero may be unrealistic.
3. Handle selected missing values using an appropriate imputation technique.
4. Calculate the correlation matrix for numerical variables.
5. Visualize correlations using a heatmap.
6. Calculate Euclidean, Manhattan, and Cosine distances for selected observations.
7. Apply an appropriate normalization or standardization technique.
8. Compare the data before and after preprocessing.
9. Interpret the obtained correlation and distance measures.

Description of the Experiment

Correlation analysis helps determine the strength and direction of relationships between variables.

For example, the relationship between:

Glucose and BMI

can be analyzed using correlation.

Distance measures determine the similarity or dissimilarity between observations.

Data preprocessing is performed to improve the quality and consistency of data before further statistical or machine learning analysis.

The experiment follows: Raw Dataset, Identify Invalid/Missing Values, Data Preprocessing, Correlation Analysis, Distance Calculation and Interpretation

Detailed Description of Statistical Methods

Correlation

Correlation measures the strength and direction of the linear relationship between two variables.

The Pearson correlation coefficient is:



$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

The value of r lies between:

$$-1 \leq r \leq 1$$

$r \approx 1$: Strong positive relationship

$r \approx -1$: Strong negative relationship

$r \approx 0$: Weak or no linear relationship

A correlation heatmap can be used to visualize relationships among multiple variables.

Euclidean Distance

Euclidean distance between two observations is:

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

It represents the straight-line distance between two points.

Manhattan Distance

Manhattan distance is:

$$d(x, y) = \sum_{i=1}^n |x_i - y_i|$$

It measures distance by summing the absolute differences between corresponding features.

Cosine Distance

Cosine similarity is:

$$\cos(\theta) = \frac{x \cdot y}{\|x\| \|y\|}$$

Cosine distance is:

$$d_{\cosine} = 1 - \cos(\theta)$$

It measures the difference in the direction of two vectors.

Data Preprocessing

Missing Value Handling

Invalid or missing values may be replaced using techniques such as:

- Mean imputation
- Median imputation

For medical variables, median imputation may be preferred when extreme values are present.



Normalization

Min-Max normalization transforms values into a specified range, usually [1]:

$$x' = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Standardization

Standardization transforms a variable to have mean 0 and standard deviation 1:

$$Z = \frac{x - \mu}{\sigma}$$

Scaling is important when distance-based methods are used because variables with larger numerical ranges can otherwise dominate the distance calculation.

Output :

```
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 4 Filter (e.g. text, includ...
[Running] python -u "d:\Statistics\exp3.py"

0      6      148      72 ...      0.627      50      1
1      1      85      66 ...      0.351      31      0
2      8      183      64 ...      0.672      32      1
3      1      89      66 ...      0.167      21      0
4      0      137      40 ...      2.288      33      1

[5 rows x 9 columns]
Pregnancies      0
Glucose           0
BloodPressure     0
SkinThickness     0
Insulin           0
BMI              0
DiabetesPedigreeFunction  0
Age              0
Outcome          0
dtype: int64
Glucose           0
BloodPressure     0
SkinThickness     0
Insulin           0
BMI              0
DiabetesPedigreeFunction  0
Age              0
Outcome          0
dtype: int64
Pregnancies      0
Glucose           0
BloodPressure     0
SkinThickness     0
Insulin           0
BMI              0
DiabetesPedigreeFunction  0
Age              0
Outcome          0
dtype: int64
Pregnancies      0
```



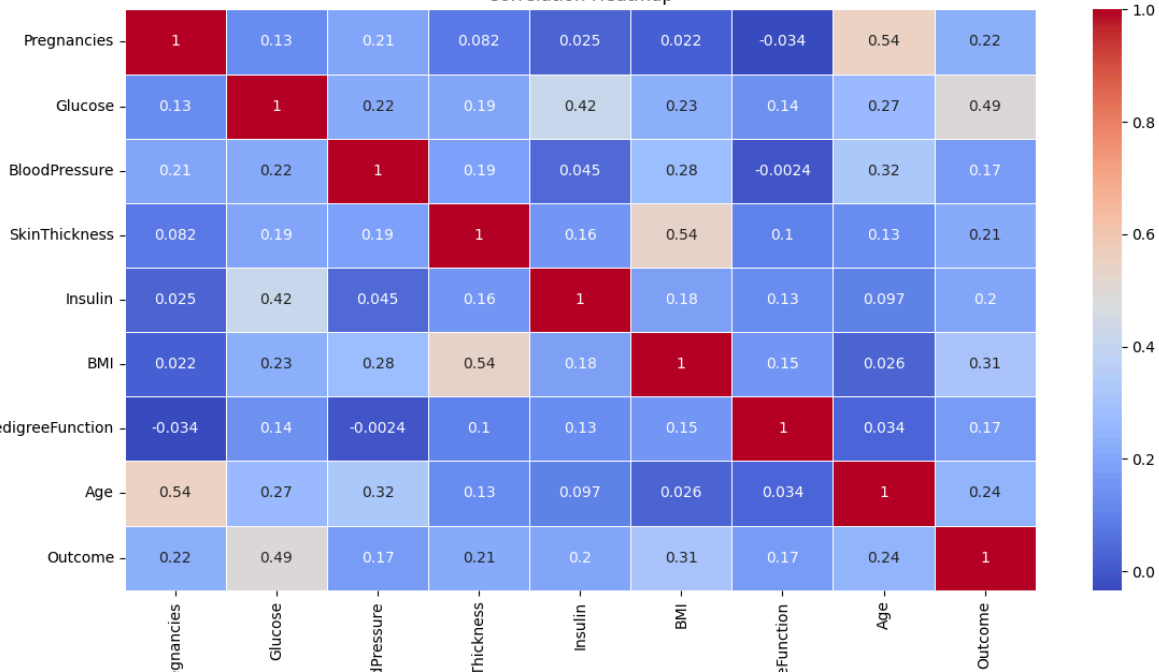
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```
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 4 Filter (e.g. text, exclu
dtype: int64
Pregnancies      0
Glucose          5
BloodPressure    35
SkinThickness    227
Insulin         374
BMI             11
DiabetesPedigreeFunction  0
Age             0
Outcome         0
dtype: int64
Pregnancies      0
Glucose          0
BloodPressure    0
SkinThickness    0
Insulin          0
BMI             0
DiabetesPedigreeFunction  0
Age             0
Outcome         0
dtype: int64
Pregnancies      0.128213 0.128213 ... 0.544341 0.221898
Glucose          0.128213 1.000000 ... 0.266909 0.492782
BloodPressure    0.208615 0.218937 ... 0.324915 0.165723
SkinThickness    0.081770 0.192615 ... 0.126107 0.214873
Insulin          0.025047 0.419451 ... 0.097101 0.203790
BMI             0.021559 0.231049 ... 0.025597 0.312038
DiabetesPedigreeFunction -0.033523 0.137327 ... 0.033561 0.173844
Age             0.544341 0.266909 ... 1.000000 0.238356
Outcome         0.221898 0.492782 ... 0.238356 1.000000

[9 rows x 9 columns]
Pregnancies  Glucose  BloodPressure  ...  BMI  DiabetesPedigreeFunction  Age
0           6    148.0         72.0  ...  33.6             0.627      50
1           1     85.0         66.0  ...  26.6             0.351      31
```

Correlation Heatmap





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```
PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 4 Filter (e.g. text, includ...)

[2 rows x 8 columns]
Euclidean Distance
[[ 0.        66.90348403]
 [66.90348403  0.        ]]

Manhattan Distance
[[ 0.        106.276]
 [106.276  0.        ]]

Cosine Distance
[[0.        0.03162267]
 [0.03162267  0.        ]]

Pregnancies  Glucose  ...  DiabetesPedigreeFunction  Age
0      0.639947  0.866045  ...                0.468492  1.425995
1     -0.844885 -1.205066  ...               -0.365061 -0.190672
2      1.233880  2.016662  ...                0.604397 -0.105584
3     -0.844885 -1.073567  ...               -0.920763 -1.041549
4     -1.141852  0.504422  ...                5.484909 -0.020496

[5 rows x 8 columns]
Pregnancies  Glucose  BloodPressure  ...  BMI  DiabetesPedigreeFunction  Age
0           6      148.0           72.0  ...   33.6                0.627  50
1           1       85.0           66.0  ...   26.6                0.351  31
2           8      183.0           64.0  ...   23.3                0.672  32
3           1       89.0           66.0  ...   28.1                0.167  21
4           0      137.0           40.0  ...   43.1                2.288  33

[5 rows x 8 columns]
Pregnancies  Glucose  ...  DiabetesPedigreeFunction  Age
0      0.639947  0.866045  ...                0.468492  1.425995
1     -0.844885 -1.205066  ...               -0.365061 -0.190672
2      1.233880  2.016662  ...                0.604397 -0.105584
3     -0.844885 -1.073567  ...               -0.920763 -1.041549
4     -1.141852  0.504422  ...                5.484909 -0.020496
```

```
[5 rows x 8 columns]
Pregnancies  Glucose  ...  Age  Outcome
count  768.000000  768.000000  ...  768.000000  768.000000
mean    3.845052  121.656250  ...  33.240885  0.348958
std     3.369578  30.438286  ...  11.760232  0.476951
min     0.000000  44.000000  ...  21.000000  0.000000
25%     1.000000  99.750000  ...  24.000000  0.000000
50%     3.000000  117.000000  ...  29.000000  0.000000
75%     6.000000  140.250000  ...  41.000000  1.000000
max    17.000000  199.000000  ...  81.000000  1.000000

[8 rows x 9 columns]
Pregnancies  Glucose  ...  DiabetesPedigreeFunction  Age
count  7.680000e+02  7.680000e+02  ...  7.680000e+02  7.680000e+02
mean   -6.476301e-17  4.625929e-18  ...  2.451743e-16  1.931325e-16
std     1.000652e+00  1.000652e+00  ...  1.000652e+00  1.000652e+00
min    -1.141852e+00 -2.552931e+00  ...  -1.189553e+00 -1.041549e+00
25%    -8.448851e-01 -7.201630e-01  ...  -6.889685e-01 -7.862862e-01
50%    -2.509521e-01 -1.530732e-01  ...  -3.001282e-01 -3.608474e-01
75%     6.399473e-01  6.112653e-01  ...  4.662269e-01  6.602056e-01
max     3.906578e+00  2.542658e+00  ...  5.883565e+00  4.063716e+00

[8 rows x 8 columns]
```



Conclusion

Correlation analysis, distance measures, and data preprocessing techniques were implemented on the Pima Indians Diabetes Dataset. Correlation analysis helped identify relationships between medical variables, while distance measures quantified the similarity or dissimilarity between observations. Data preprocessing techniques helped improve the consistency and suitability of the data for further analysis and machine learning applications.

Questions to be Answered by Students

1. Which pair of variables has the strongest positive correlation in the dataset?

ANS : The strongest positive correlation between two independent variables is:

1. Pregnancies and Age = 0.54
2. SkinThickness and BMI = 0.54
3. Both pairs have the highest positive correlation value of 0.54.

2. Which pair of variables has the strongest negative correlation, if any?

ANS : The strongest negative correlation is:

- Pregnancies and DiabetesPedigreeFunction = -0.034
- This value is very close to zero, indicating there is no significant negative correlation between any pair of variables in the dataset.

3. How do the Euclidean, Manhattan, and Cosine distances differ for the selected observations?

ANS : For the first two observations:

- Euclidean Distance = 66.903
 - Measures the straight-line distance between the two observations.
- Manhattan Distance = 106.276
 - Measures the sum of the absolute differences across all features.
 - It is larger because it adds the differences in each dimension.
- Cosine Distance = 0.0316
 - Measures the difference in the direction (angle) of the feature vectors rather than their magnitude.
 - The small value indicates that the two observations have similar patterns despite differences in their actual values.



4. Which preprocessing technique was applied to the invalid or missing values, and why was it selected?

ANS : The preprocessing technique used was Median Imputation.

- Invalid zero values in Glucose, BloodPressure, SkinThickness, Insulin, and BMI were replaced with NaN.
- These missing values were then filled using the median of each respective column.
- The median was chosen because it is less affected by outliers than the mean, making it suitable for medical data.

5. How does normalization or standardization affect the calculated distance between observations?

ANS : Standardization was applied using StandardScaler.

Effects:

- It transforms each feature to have a mean of 0 and a standard deviation of 1.
- Features with larger numerical ranges no longer dominate the distance calculations.
- All variables contribute equally when computing Euclidean or Manhattan distances.
- This results in more reliable and meaningful comparisons between observations, especially for machine learning algorithms.